

LINUX FUNDAMENTALS | PROJECT: INFO EXTRACTOR

UNIT: RW-University-II

CODE: s10

System Diagnostics

1. Identify the system's public IP.

```
(kali㉿kali)-[~]  
$ curl ifconfig.co  
41.186.1.30
```

2. Identify the private IP address assigned to the system's network interface.

```
(kali㉿kali)-[~]  
$ ifconfig | grep broadcast | awk '{print $2}'  
192.168.1.24
```

3. Display the MAC address (masking sensitive portions for security).

```
(kali㉿kali)-[~]  
$ ifconfig | grep ether | awk '{print $2}'  
08:00:27:xx:xx:f5
```

4. Display the percentage of CPU usage for the top 5 processes.

```
(kali㉿kali)-[~]  
$ ps aux --sort=-%cpu | head -n 6 | awk '{print $3,$11}' | sed '1s/^/CPU(%) Command\n/'  
  
CPU(%) Command  
%CPU COMMAND  
5.3 /usr/lib/firefox-esr/firefox-esr  
1.7 /usr/lib/xorg/Xorg  
0.3 /usr/bin/VBoxClient  
0.2 /usr/lib/firefox-esr/firefox-esr  
0.1 /usr/bin/containerd
```

5. Display memory usage statistics: total and available memory.

```
(kali㉿kali)-[~]  
$ free -h --mega  
              total        used        free      shared  buff/cache   available  
Mem:          5.5G         1.7G         2.5G          32M         1.7G         3.8G  
Swap:          1.1G           0B         1.1G
```

6. List active system services with their status.

```
(kali㉿kali)-[~]  
$ service --status-all | grep '+'  
[ + ] cron  
[ + ] dbus  
[ + ] haveged  
[ + ] kmod  
[ + ] lightdm  
[ + ] networking
```

7. Locate the Top 10 Largest Files in /home.

```
(kali㉿kali)-[~]  
$ du /home | sort -n | tail -n 10  
1254284 /home/kali/.cache  
1412836 /home/kali/Downloads/burp/burpsuite_pro_v2023.12.1.2  
2115268 /home/kali/Downloads/burp  
2714724 /home/kali/Downloads  
8876804 /home/kali/.local/share/Trash/files  
8876920 /home/kali/.local/share/Trash  
8877164 /home/kali/.local/share  
9008616 /home/kali/.local  
15155196 /home/kali  
15155200 /home
```